

Gravitational Waves

Editted by Yan Qian-Bo

August 4, 2026

1 Linearized gravity

The gravity can be linearized by

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

where $|h_{\mu\nu}| \ll 1$ and $|\partial_\sigma h_{\mu\nu}| \ll 1$

Then, we define the inverse metric

$$g^{\mu\nu} = \eta^{\mu\nu} + k^{\mu\nu}$$

with the identity $g_{\mu\sigma}g^{\sigma\nu} = \delta_\mu^\nu$

$$\begin{aligned}\delta_\mu^\nu &= g_{\mu\sigma}g^{\sigma\nu} = (\eta_{\mu\sigma} + h_{\mu\sigma})(\eta^{\sigma\nu} + k^{\sigma\nu}) \\ &\approx \delta_\mu^\nu + \eta_{\mu\sigma}k^{\sigma\nu} + h_{\mu\sigma}\eta^{\sigma\nu}\end{aligned}$$

$$\eta_{\mu\sigma}k^{\sigma\nu} = -h_{\mu\sigma}\eta^{\sigma\nu} \quad \Rightarrow \quad \eta^{\rho\mu}\eta_{\mu\sigma}k^{\sigma\nu} = -h_{\mu\sigma}\eta^{\rho\mu}\eta^{\sigma\nu} \quad \Rightarrow \quad k^{\rho\nu} = -h_{\mu\sigma}\eta^{\rho\mu}\eta^{\sigma\nu} = -h^{\rho\nu}$$

The Christoffel symbols is given by

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2}g^{\sigma\alpha}(\partial_\mu g_{\nu\alpha} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu}) \approx \frac{1}{2}\eta^{\sigma\alpha}(\partial_\mu h_{\nu\alpha} + \partial_\nu h_{\mu\alpha} - \partial_\alpha h_{\mu\nu})$$

The Riemann tensor is

$$\begin{aligned}R^\rho_{\sigma\mu\nu} &= \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\nu\sigma}^\gamma \Gamma_{\mu\gamma}^\rho - \Gamma_{\mu\sigma}^\gamma \Gamma_{\nu\gamma}^\rho \\ &\approx \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho \\ &= \frac{1}{2}\eta^{\rho\alpha}(\partial_\mu \partial_\nu h_{\alpha\sigma} + \partial_\mu \partial_\sigma h_{\alpha\nu} - \partial_\mu \partial_\alpha h_{\nu\sigma} - \partial_\nu \partial_\mu h_{\alpha\sigma} - \partial_\nu \partial_\sigma h_{\alpha\mu} + \partial_\nu \partial_\alpha h_{\mu\sigma}) \\ &= \frac{1}{2}\eta^{\rho\alpha}(\partial_\mu \partial_\sigma h_{\alpha\nu} - \partial_\mu \partial_\alpha h_{\nu\sigma} - \partial_\nu \partial_\sigma h_{\alpha\mu} + \partial_\nu \partial_\alpha h_{\mu\sigma})\end{aligned}$$

Then, the Ricci tensor is

$$\begin{aligned} R_{\sigma\nu} &= R^\mu_{\sigma\mu\nu} = \frac{1}{2}\eta^{\mu\alpha}(\partial_\mu\partial_\sigma h_{\alpha\nu} - \partial_\mu\partial_\alpha h_{\nu\sigma} - \partial_\nu\partial_\sigma h_{\alpha\mu} + \partial_\nu\partial_\alpha h_{\mu\sigma}) \\ &= \frac{1}{2}(\partial_\mu\partial_\sigma h^\mu_{\nu} - \square h_{\nu\sigma} - \partial_\nu\partial_\sigma h + \partial_\nu\partial^\mu h_{\mu\sigma}) \end{aligned}$$

where we define $\square = \partial_\mu\partial^\mu$ and $h = h^\mu_{\mu} = \eta_{\mu\nu}h^{\mu\nu}$.

The Ricci scalar is

$$\begin{aligned} R &= \eta^{\sigma\nu}R_{\sigma\nu} = \frac{1}{2}\eta^{\sigma\nu}(\partial_\mu\partial_\sigma h^\mu_{\nu} - \square h_{\nu\sigma} - \partial_\nu\partial_\sigma h + \partial_\nu\partial^\mu h_{\mu\sigma}) \\ &= \partial_\mu\partial_\sigma h^{\mu\sigma} - \square h \end{aligned}$$

The Einstein tensor is

$$\begin{aligned} G_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \approx \frac{1}{2}(\partial_\sigma\partial_\mu h^\sigma_{\nu} - \square h_{\mu\nu} - \partial_\nu\partial_\mu h + \partial_\nu\partial^\sigma h_{\mu\sigma}) - \frac{1}{2}\eta_{\mu\nu}(\partial_\rho\partial_\sigma h^{\rho\sigma} - \square h) \\ &= \frac{1}{2}(\partial^\sigma\partial_\mu h_{\sigma\nu} - \square h_{\mu\nu} - \partial_\mu\partial_\nu h + \partial^\sigma\partial_\nu h_{\mu\sigma}) - \frac{1}{2}\eta_{\mu\nu}(\partial^\sigma\partial^\rho h_{\sigma\rho} - \square h) \end{aligned}$$

Now, if we define

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h \quad \Rightarrow \quad h_{\mu\nu} = \bar{h}_{\mu\nu} + \frac{1}{2}\eta_{\mu\nu}h$$

the Einstein tensor becomes

$$G_{\mu\nu} = \frac{1}{2}(\partial^\sigma\partial_\mu \bar{h}_{\sigma\nu} - \partial^\sigma\partial_\sigma \bar{h}_{\mu\nu} + \partial^\sigma\partial_\nu \bar{h}_{\mu\sigma} - \eta_{\mu\nu}\partial^\sigma\partial^\rho \bar{h}_{\sigma\rho})$$

We are ready to solve the Einstein equation for the linearized gravity.

$$G_{\mu\nu} = \frac{1}{2}(\partial^\sigma\partial_\mu \bar{h}_{\sigma\nu} - \partial^\sigma\partial_\sigma \bar{h}_{\mu\nu} + \partial^\sigma\partial_\nu \bar{h}_{\mu\sigma} - \eta_{\mu\nu}\partial^\sigma\partial^\rho \bar{h}_{\sigma\rho}) = \frac{8\pi G}{c^4}T_{\mu\nu}$$

2 Lorentz gauge

If we choose the Lorentz gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$, then the Einstein equation becomes

$$G_{\mu\nu} = -\frac{1}{2}\partial^\sigma\partial_\sigma \bar{h}_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$$

To achieve Lorentz gauge, we consider a coordinate transformation

$$\tilde{x}^\mu = x^\mu + \xi^\mu$$

where ξ^μ is a displacement field with $\left|\frac{\partial\xi^\mu}{\partial x^\nu}\right|^2 \ll 1$ and $\left|\frac{\partial\xi^\mu}{\partial \tilde{x}^\nu}\right|^2 \ll 1$.

The transformation can be written as

$$\begin{aligned}
\frac{\partial \tilde{x}^\mu}{\partial x^\nu} &= \frac{\partial x^\mu}{\partial x^\nu} + \frac{\partial \xi^\mu}{\partial x^\nu} = \delta^\mu{}_\nu + \frac{\partial \xi^\mu}{\partial x^\nu} \\
\frac{\partial x^\mu}{\partial \tilde{x}^\nu} &= \frac{\partial \tilde{x}^\mu}{\partial \tilde{x}^\nu} - \frac{\partial \xi^\mu}{\partial x^\sigma} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} \\
&= \delta^\mu{}_\nu - \frac{\partial \xi^\mu}{\partial x^\sigma} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} \\
&= \delta^\mu{}_\nu - \frac{\partial \xi^\mu}{\partial x^\sigma} \left(\delta^\sigma{}_\nu - \frac{\partial \xi^\sigma}{\partial \tilde{x}^\nu} \right) \\
&\approx \delta^\mu{}_\nu - \frac{\partial \xi^\mu}{\partial x^\nu}
\end{aligned}$$

The metric for the new coordinate is

$$\begin{aligned}
\tilde{g}_{\mu\nu} &= \tilde{e}_\mu \cdot \tilde{e}_\nu \\
&= \frac{\partial}{\partial \tilde{x}^\mu} \cdot \frac{\partial}{\partial \tilde{x}^\nu} \\
&= \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial}{\partial x^\alpha} \cdot \frac{\partial x^\beta}{\partial \tilde{x}^\nu} \frac{\partial}{\partial x^\beta} \\
&= \left(\delta^\alpha{}_\mu - \frac{\partial \xi^\alpha}{\partial x^\mu} \right) \left(\delta^\beta{}_\nu - \frac{\partial \xi^\beta}{\partial x^\nu} \right) g_{\alpha\beta} \\
&\approx \left(\delta^\alpha{}_\mu \delta^\beta{}_\nu - \delta^\beta{}_\nu \frac{\partial \xi^\alpha}{\partial x^\mu} - \delta^\alpha{}_\mu \frac{\partial \xi^\beta}{\partial x^\nu} \right) (\eta_{\alpha\beta} + h_{\alpha\beta}) \\
&\approx \eta_{\mu\nu} + h_{\mu\nu} - \eta_{\alpha\nu} \frac{\partial \xi^\alpha}{\partial x^\mu} - \eta_{\mu\beta} \frac{\partial \xi^\beta}{\partial x^\nu} \\
&= g_{\mu\nu} - \frac{\partial \xi_\nu}{\partial x^\mu} - \frac{\partial \xi_\mu}{\partial x^\nu}
\end{aligned}$$

Writing $\tilde{g}_{\mu\nu} = \eta_{\mu\nu} + \tilde{h}_{\mu\nu}$, we see

$$\tilde{h}_{\mu\nu} = h_{\mu\nu} - \frac{\partial \xi_\nu}{\partial x^\mu} - \frac{\partial \xi_\mu}{\partial x^\nu}$$

Noticing that

$$\frac{\partial \xi_\mu}{\partial \tilde{x}^\nu} = \frac{\partial \xi_\mu}{\partial x^\sigma} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} = \frac{\partial \xi_\mu}{\partial x^\sigma} \left(\delta^\sigma{}_\nu - \frac{\partial \xi^\sigma}{\partial x^\nu} \right) \approx \frac{\partial \xi_\mu}{\partial x^\nu}$$

and that

$$\begin{aligned}
\frac{\partial \tilde{h}_{\mu\nu}}{\partial \tilde{x}^\alpha} &= \frac{\partial h_{\mu\nu}}{\partial \tilde{x}^\alpha} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\nu}{\partial x^\mu} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\mu}{\partial x^\nu} \\
&= \frac{\partial x^\sigma}{\partial \tilde{x}^\alpha} \frac{\partial h_{\mu\nu}}{\partial x^\sigma} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\nu}{\partial x^\mu} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\mu}{\partial x^\nu} \\
&= \left(\delta^\sigma{}_\alpha - \frac{\partial \xi^\sigma}{\partial x^\alpha} \right) \frac{\partial h_{\mu\nu}}{\partial x^\sigma} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\nu}{\partial x^\mu} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\mu}{\partial x^\nu} \\
&\approx \frac{\partial h_{\mu\nu}}{\partial x^\alpha} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\nu}{\partial x^\mu} - \frac{\partial}{\partial x^\alpha} \frac{\partial \xi_\mu}{\partial x^\nu}
\end{aligned}$$

so we can just write

$$\partial_\alpha \tilde{h}_{\mu\nu} = \partial_\alpha h_{\mu\nu} - \partial_\alpha \partial_\mu \xi_\nu - \partial_\alpha \partial_\nu \xi_\mu$$

The Ricci tensor does not change under the small displacement transformation

$$\begin{aligned} \tilde{R}^\rho_{\sigma\mu\nu} &= \frac{1}{2} \eta^{\rho\alpha} \left(\partial_\mu \partial_\sigma \tilde{h}_{\alpha\nu} - \partial_\mu \partial_\alpha \tilde{h}_{\nu\sigma} - \partial_\nu \partial_\sigma \tilde{h}_{\alpha\mu} + \partial_\nu \partial_\alpha \tilde{h}_{\mu\sigma} \right) \\ &= \frac{1}{2} \eta^{\rho\alpha} (\partial_\mu \partial_\sigma h_{\alpha\nu} - \partial_\mu \partial_\sigma \partial_\nu \xi_\alpha - \partial_\mu \partial_\sigma \partial_\alpha \xi_\nu - \partial_\mu \partial_\alpha h_{\nu\sigma} + \partial_\mu \partial_\alpha \partial_\nu \xi_\sigma + \partial_\mu \partial_\alpha \partial_\sigma \xi_\nu \\ &\quad - \partial_\nu \partial_\sigma h_{\alpha\mu} + \partial_\nu \partial_\sigma \partial_\alpha \xi_\mu + \partial_\nu \partial_\sigma \partial_\mu \xi_\alpha + \partial_\nu \partial_\alpha h_{\mu\sigma} - \partial_\nu \partial_\alpha \partial_\mu \xi_\sigma - \partial_\nu \partial_\alpha \partial_\sigma \xi_\mu) \\ &= \frac{1}{2} \eta^{\rho\alpha} (\partial_\mu \partial_\sigma h_{\alpha\nu} - \partial_\mu \partial_\alpha h_{\nu\sigma} - \partial_\nu \partial_\sigma h_{\alpha\mu} + \partial_\nu \partial_\alpha h_{\mu\sigma}) \\ &= R^\rho_{\sigma\mu\nu} \end{aligned}$$

Next, we see

$$\begin{aligned} \tilde{\bar{h}}_{\mu\nu} &= \tilde{h}_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \tilde{h} \\ &= h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu - \frac{1}{2} \eta_{\mu\nu} \eta^{\alpha\beta} (h_{\alpha\beta} - \partial_\alpha \xi_\beta - \partial_\beta \xi_\alpha) \\ &= \bar{h}_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu + \eta_{\mu\nu} \partial_\alpha \xi^\alpha \end{aligned}$$

The Lorentz gauge takes the condition

$$\begin{aligned} \partial_\mu \tilde{\bar{h}}^{\mu\nu} &= \partial_\mu (\bar{h}^{\mu\nu} - \partial^\mu \xi^\nu - \partial^\nu \xi^\mu + \eta^{\mu\nu} \partial_\alpha \xi^\alpha) \\ &= \partial_\mu \bar{h}^{\mu\nu} - \square \xi^\nu - \partial_\mu \partial^\nu \xi^\mu + \partial^\nu \partial_\alpha \xi^\alpha \\ &= \partial_\mu \bar{h}^{\mu\nu} - \square \xi^\nu \end{aligned}$$

Hence, if we choose $\square \xi^\nu = \partial_\mu \bar{h}^{\mu\nu}$, then we have $\partial_\mu \tilde{\bar{h}}^{\mu\nu} = 0$.

Now, if we solve the Einstein equation in vacuum $T_{\mu\nu} = 0$, we only need to solve the wave equation

$$\square \bar{h}_{\mu\nu} = 0 \quad \Rightarrow \quad \frac{1}{c^2} \frac{\partial^2 \bar{h}_{\mu\nu}}{\partial t^2} = \nabla^2 \bar{h}_{\mu\nu}$$

3 Transverse and Traceless Gauge

The solution of the wave equation $\square \bar{h}_{\mu\nu}$ is

$$\bar{h}_{\mu\nu} = A_{\mu\nu} \cos(\omega t - \mathbf{k} \cdot \mathbf{x})$$

where $A_{\mu\nu} = A_{\nu\mu}$ is symmetric and $\omega = c|\mathbf{k}|$

Defining the wave vector $k_\sigma = \left(\frac{\omega}{c}, -\mathbf{k}\right)$, we can also write

$$\bar{h}_{\mu\nu} = \text{Re} \left(A_{\mu\nu} e^{ik_\sigma x^\sigma} \right)$$

and the wave vector is light-like $k_\sigma k^\sigma = 0$.

To see the polarization of the gravitational waves, we reduce the 16 components in $A_{\mu\nu}$.

- Since $A_{\mu\nu} = A_{\nu\mu}$, there is only 10 independent components.
- Lorentz gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$ reduces $A_{\mu\nu}$ to 6 independent components.

$$\partial_\mu \bar{h}^{\mu\nu} = 0 \quad \Rightarrow \quad k_\mu A^{\mu\nu} = 0$$

- We can also choose a specific Lorentz gauge by a transformation

$$\tilde{x}^\mu = x^\mu + \xi^\mu \quad \Rightarrow \quad \tilde{\bar{h}}_{\mu\nu} = \bar{h}_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu - \eta_{\mu\nu} \partial_\sigma \xi^\sigma \quad \Rightarrow \quad \partial^\mu \tilde{\bar{h}}_{\mu\nu} = \partial^\mu \bar{h}_{\mu\nu} - \square \xi_\nu$$

where $\square \xi_\nu = 0$ so the Lorentz gauge is still satisfied. The transverse-traceless gauge defines 4 constraints on $A_{\mu\nu}$. Hence, there are 2 independent components left in $A_{\mu\nu}$.

The transverse gauge $U^\mu A_{\mu\nu} = 0$ gives 3 constraints because the rank of the matrix $A_{\mu\nu}$ is at most 3 due to the Lorentz gauge $k^\mu A_{\mu\nu} = 0$. Without loss of generality, we can set

$$U^\mu = (c, 0, 0, 0) \quad \text{and} \quad k^\mu = \left(\frac{\omega}{c}, 0, 0, \frac{\omega}{c}\right) \quad \Rightarrow \quad A_{t\nu} = A_{z\nu} = 0$$

which represents that the gravitational wave propagates in z-direction in the rest frame.

The traceless gauge gives 1 constraint.

$$A^\mu{}_\mu = \eta^{\mu\nu} A_{\mu\nu} = A_{tt} - A_{xx} - A_{yy} - A_{zz} = 0$$

Also, we see

$$0 = \eta^{\mu\nu} \bar{h}_{\mu\nu} = \eta^{\mu\nu} \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) = h - 2h = -h \quad \Rightarrow \quad h = 0 \quad \Rightarrow \quad \bar{h}_{\mu\nu} = h_{\mu\nu}$$

Now, with the above constraints, the matrix form of $A_{\mu\nu}$ is now reduced to

$$A_{\mu\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & A_{xx} & A_{xy} & 0 \\ 0 & A_{xy} & -A_{xx} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = A_+ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + A_\times \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Here, we call $A_{xx} = A_+$ and $A_{xy} = A_\times$ the amplitude of the plus and cross polarizations.

4 Effect of Gravitational Waves on Free Particles

Consider the geodesic equation with $U^\mu = \frac{dx^\mu}{d\tau}$

$$\frac{d^2x^\sigma}{d\tau^2} + \Gamma_{\mu\nu}^\sigma \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0 \quad \Rightarrow \quad \frac{dU^\sigma}{d\tau} + \Gamma_{\mu\nu}^\sigma U^\mu U^\nu = 0$$

It is easy to see the constant $U^\sigma = (c, 0, 0, 0)$ satisfies the equation since $h_{t\nu} = 0$.

$$\begin{cases} \frac{dU^\sigma}{d\tau} = 0 \\ \Gamma_{\mu\nu}^\sigma U^\mu U^\nu = c^2 \Gamma_{tt}^\sigma = \frac{1}{2} c^2 \eta^{\sigma\alpha} (\partial_t h_{t\alpha} + \partial_t h_{t\alpha} - \partial_\alpha h_{tt}) = 0 \end{cases} \quad \Rightarrow \quad \frac{dU^\sigma}{d\tau} + \Gamma_{\mu\nu}^\sigma U^\mu U^\nu = 0$$

Hence, with a starting point b^μ , a simple solution is

$$x^\mu = b^\mu + U^\mu \tau = (c\tau + b^t, b^x, b^y, b^z)$$

Consider a vector $v^\mu = (0, v_x, v_y, 0)$ on the xy-plane, the length of the vector changes with time.

$$\begin{aligned} v_\mu v^\mu &= g_{\mu\nu} v^\mu v^\nu \\ &= (\eta_{\mu\nu} + h_{\mu\nu}) v^\mu v^\nu \\ &= v_x^2 (-1 + A_+ e^{ik_z(x-ct)}) + 2v_x v_y A_\times e^{ik_z(x-ct)} + v_y^2 (-1 - A_+ e^{ik_z(x-ct)}) \\ &= -(v_x^2 + v_y^2) + [(v_x^2 - v_y^2) A_+ + 2v_x v_y A_\times] e^{ik_z(x-ct)} \end{aligned}$$

We see that

$$-(v_x^2 + v_y^2) = -L^2$$

represents the squared Minkowski norm of the vector, while the perturbation

$$\Delta = [(v_x^2 - v_y^2) A_+ + 2v_x v_y A_\times] e^{ik_z(x-ct)}$$

represents the contribution induced by the gravitational wave.

- For a vector in x-direction $v_y = 0$, we see

$$\Delta = v_x^2 A_+ e^{ik_z(x-ct)}$$

For a vector in y-direction $v_x = 0$, we see

$$\Delta = -v_y^2 A_+ e^{ik_z(x-ct)} = v_y^2 A_+ e^{i[k_z(x-ct) + \pi]}$$

This is the **plus mode** with the amplitude characterized by A_+ .

- For a vector in diagonal $v_y = v_x$, we see

$$\Delta = 2v_x^2 A_{\times} e^{ik_z(x-ct)}$$

For a vector in the other diagonal $v_y = -v_x$, we see

$$\Delta = -2v_x^2 A_{\times} e^{ik_z(x-ct)} = 2v_x^2 A_{\times} e^{i[k_z(x-ct)+\pi]}$$

This is the **cross mode** with the amplitude characterized by $2A_{\times}$.